

How Social Media Peer Comments Influence Privacy Decisions in Photo Sharing: Context and Individual Differences Cause Comments to Backfire

Tangila Islam Tanni
University of Central Florida
Department of Computer Science
Orlando, Florida, United States
tangilaislam.tanni@ucf.edu

Yan Solihin
University of Central Florida
Department of Computer Science
Orlando, Florida, United States
yan.solihin@ucf.edu

Renita Washburn
University of Central Florida
School of Modeling, Simulation and Training
Orlando, Florida, United States
renitawashburn@ucf.edu

Mary Jean Amon
Indiana University Bloomington
Department of Informatics
Bloomington, Indiana, United States
mjamon@iu.edu

Abstract

Although extensive research has examined social media nudges to influence privacy, comparably less work investigates mechanisms of peer-to-peer influence, or the ways in which peer groups can amplify or dampen privacy considerations. This study takes an experimental approach to examining how people are influenced to share potentially sensitive photos of others without consent—a common practice on social media—by comparing the effects of encouraging, discouraging, or mixed feedback social media comments. The results show that discouraging peer comments reduce the sharing of others' photos when content portrayed targets negatively. However, peer influence backfired among users with a history of online misconduct, with both encouraging and discouraging comments leading to more photo sharing. Moreover, a mix of comments that both discouraged and encouraged sharing led to the highest levels of sharing. Linking to prior work, this research establishes a broader understanding of when privacy nudges are likely to backfire on social media: for users with past online misconduct and when information goes against social norms. The findings advance understanding of how peer-to-peer dynamics shape online information flows and highlight the complex effects of Web-based social feedback on privacy decisions. These insights provide actionable design recommendations for platforms to foster responsible, privacy-conscious sharing norms.

CCS Concepts

• **Security and privacy** → **Social aspects of security and privacy**.

Keywords

Photo Sharing; Privacy; Nudge; Social Media; Peer Influence

ACM Reference Format:

Tangila Islam Tanni, Renita Washburn, Yan Solihin, and Mary Jean Amon. 2026. How Social Media Peer Comments Influence Privacy Decisions in Photo Sharing: Context and Individual Differences Cause Comments to Backfire. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792468>

1 Introduction

The Web has transformed how individuals communicate, collaborate, and express themselves, with social media platforms emerging as a dominant layer of Web-based interaction [44]. Platforms such as Facebook [68], Instagram [62], and Snapchat [49] exemplify the Web's role in enabling large-scale photo sharing [44, 49], often in real time [19, 35]. While this infrastructure has democratized self-expression and accelerated content diffusion, it has also created new challenges for privacy [56, 67], as personal information of individuals depicted in shared content can be exposed beyond their control [65]. Understanding, evaluating, and improving such socio-technical dynamics is a central concern of the Web research community. In particular, it is important to understand how quick sharing decisions—that may impact other people's privacy—are made in social contexts.

When users post photos of other people on social media, they often do so without permission or considering the potential privacy implications for those featured in the content [33]. This oversight can lead to 'interdependent privacy' (IDP) violations, which occur when individuals share potentially sensitive information about others without their consent [13, 72]. For instance, imagine Alex and Janet attend a private gathering together. Alex takes a photo of Janet at the event and later shares it on social media without asking Janet's permission. Janet is in the early stages of pregnancy and the photo incorrectly makes it appear as though she might have been drinking alcohol. After revealing her pregnancy to friends, the photo starts drawing concern from others. Janet is mortified to learn that, not only are acquaintances circulating the photo in private chats, but the photo was also posted in an online gossip forum. Such incidents highlight how Web-based interactions create



This work is licensed under a Creative Commons Attribution 4.0 International License. *WWW '26, Dubai, United Arab Emirates*.

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2307-0/2026/04

<https://doi.org/10.1145/3774904.3792468>

complex IDP risks, where one person's actions online can unintentionally compromise another's privacy. While often harmless in intent, these posts can lead to serious consequences, including emotional distress [24], harassment, job loss [23], and damaged relationships [54]. These risks are not isolated to single platforms but represent broader social and ethical challenges for the Web ecosystem.

With billions of photos shared daily on social media [28], collaborative efforts are essential to mitigate privacy risks. Since users prefer IDP management strategies they can control [53], it is important to consider the ways in which users can support social media privacy by communicating their IDP knowledge and values [6]. Users naturally shape and uphold evolving privacy norms in online spaces, but not much is known about what effective peer privacy comments might look like. Nudges that remind users to consider others' privacy can sometimes backfire, increasing the sharing of others' sensitive information [7]. It follows that similar prompts from peers could have similarly counterintuitive effects.

Prior work on privacy nudges [1, 57] and educational interventions [5, 63] show that social and behavioral mechanisms can foster more responsible engagement with Web technologies [70]. Yet, little is known about how peer comments—ubiquitous features of Web platforms—shape IDP behaviors. To address this gap, we leveraged an experimental design testing how peer comments influence IDP-relevant social media sharing decisions in regard to photo-based memes¹ that depict people in more-or-less sensitive circumstances (as rated by a large group of independent coders). Six hundred participants rated their likelihood of sharing photo-based memes depicting *potential* IDP violations—photos of varying sensitivity of identifiable individuals shared without explicit consent. We focus on photo-based memes rather than photo sharing broadly because they provide a methodologically and ethically appropriate testbed for peer influence. Pilot testing indicated that uncaptioned photos of unfamiliar individuals offered insufficient context for participants to meaningfully evaluate sharing decisions, whereas memes supply necessary textual framing while preserving identifiable visual content. To understand the influence of peer comments on IDP-sharing decisions, we necessarily narrowed our focus to photo-based memes for the following reasons: (1) they reflect real-world, shareable content commonly encountered on social media; (2) contain people's identifiable information (i.e., photo of face); (3) typically lack clear consent from the individuals depicted (hereon referred to as “meme targets”); (4) photo-based memes combine images with text captions, making them interpretable to broader audiences and providing a common context for social media exchanges; and (5) since memes are necessarily shared within groups of people, which is what makes them ‘memes,’ they provide an appropriate context for testing peer comment influence.

Photo-based memes were accompanied by a peer comment that either encouraged, discouraged, or provided mixed feedback, compared to a control condition with no peer comment. This controlled experimental design is what supported the study of peer comment effects. In particular, we were interested in both the ‘direct’ effects

of peer comments on immediate meme sharing decisions and ‘carry-over’ effects during subsequent sharing decisions for memes shown without peer comments (i.e., to determine if effects persisted). We also examined whether peer comment influence varied according to differences in meme content and individual participant characteristics. Thus, this paper addresses the overarching research question: **How peer comments influence privacy-relevant sharing decisions in social media?** To explore this question in depth, we addressed the following sub-questions:

- RQ1: How do photo sharing decisions change in response to IDP-relevant peer comments?
- RQ2: To what extent do peer comments have carry-over effects that influence subsequent photo sharing?
- RQ3: How does peer comment influence vary based on social media content (i.e., differing portrayals of meme targets)?
- RQ4: How does peer comment influence vary based on individual differences (i.e., past online misconduct)?

Taken together, we offer an assessment of how peer-to-peer interactions shape privacy-related behavior on social networks, advancing understanding of the broader impact of the Web on privacy information flows. By examining how peers influence privacy-relevant sharing on social media, our work contributes to Web research on social media, privacy, and responsible online practices, with the goal of informing platform design that helps users establish and maintain privacy norms within their communities.

2 Background

IDP encompasses the complex interplay between personal privacy and the actions of others [13]. Unlike individual privacy, IDP extends beyond personal control to include a broad range of potential violations, such as the disclosure of location or medical information, drug or sexual histories, personal characteristics, and relationships [8]. A prevalent form of IDP violation involves the creation and distribution of internet memes [31, 37]. Memes are both culturally significant and highly shareable [26], serving a range of functions such as self-expression, social connection, information dissemination, marketing, and evoking emotional responses [4, 42, 61]. However, as memes circulate widely—often disseminating both accurate and inaccurate information [16, 38, 54]—the individuals depicted may feel powerless to control the spread of their photos. This loss of control raises significant privacy concerns [3, 21, 47]. Moreover, viral meme sharing can create a gap between the intended and actual audiences [1, 43]. Repeated sharing often alters the original image or caption, changing the context and portrayal of the meme target, a phenomenon known as “context collapse” [15, 55].

While some memes positively highlight achievements or relatable moments, many are satirical, emphasizing embarrassing or controversial behaviors and even conveying hateful messages [36, 74]. These practices can transform memes into tools for public shaming or ridicule, amplifying personal flaws without consent [25, 27, 71]. Therefore, photo-based memes exemplify *potential* IDP violations, as they include personally identifying information (e.g., the target's face), are widely disseminated beyond the target's control, and are typically shared without explicit consent. The viral nature of memes underscore the potential for multimedia-based IDP violations to reach audiences at scale.

¹ Memes are user-generated digital artifacts—typically combining photos and text—that function as cultural, social, or political expressions, often through humor, and gain meaning through their remixing, circulation, and adaptation across online communities [12, 58].

Peer feedback strongly shapes behavior on social media [9, 14, 46, 51], especially photo sharing [11]. Positive comments and likes reinforce posting, creating feedback loops that increase engagement and disclosure [45, 64, 75], while negative comments or low engagement foster self-censorship or withdrawal [32]. However, this apparently straight-forward effect contrasts privacy nudge research showing the potential for backfire [7, 57]. The picture grows more complicated when considering how individual differences influence sharing decisions and online exchanges. For example, users in individualistic contexts, such as the United States, tend to express stronger privacy concerns and seek greater control over their personal information [40, 69], whereas users in collectivistic cultures are generally more open, guided by norms of reciprocity [22].

If privacy-protective sharing behaviors are influenced socially by peers, then users can be empowered to share their values and privacy safeguards, as a means to effectively foster more responsible and privacy-aware practices online. However, it is possible that peer influence is shaped by individual differences, such that the same peer interventions vary in effectiveness across users and, in some cases, may even backfire depending on context.

2.1 Present study

Given concerns over social media privacy and peer influence of sharing [2], we hypothesize that peer comments will have distinct effects: encouraging comments will increase sharing, while discouraging ones will reduce it (H1), with effects persisting in subsequent sharing decisions (H2). Given that effectiveness is likely to vary across both individual differences and content characteristics [70], we further hypothesize that meme valence will moderate outcomes, with negative content producing a greater decline in sharing than positive content (H3). In addition, users will respond differently based on individual traits: peer comments will be less effective for those with histories of online misconduct (e.g., deleted or reported comments) due to desensitization to social norms (H4). Given prior research showing potential for privacy nudge backfire [7], we consider the potential for counterintuitive peer influence effects. By accounting for social media peer interactions and individual variation in response, we move beyond one-size-fits-all models to foster a more privacy-aware online environment where peer interactions support informed sharing.

3 Method

3.1 Recruitment and ethical considerations

All procedures were approved by our Institutional Review Board (IRB). Participants were recruited and compensated through Prolific's online participant panel. To be eligible, individuals needed to be at least 18 years old, residing in the United States, fluent in English, have normal or corrected-to-normal vision, and be regular social media users with an active social media account visited at least once a week. The requirement of normal or corrected-to-normal vision was to ensure that participants could view the photo-based memes presented in the *commented meme-rating task* and *meme-rating task* (no comments). Four attention checks were used, following best practices outlined in prior research [52]. The final sample consisted of 600 participants, evenly distributed across four experimental groups (150 per group: encouraging, discouraging,

mixed, and control). Table 1 provides detailed demographics of our participants.

Table 1: Summary of participants' demographics

	n (%)
Total (N)	600 (100%)
Age (M (SD))	33.2 (10.20)
Gender	
Women	348 (58.00%)
Men	226 (37.67%)
Non-binary	26 (4.33%)
Race	
White	440 (73.33%)
Black or African American	61 (10.16%)
Asian	33 (5.50%)
Multiracial	41 (6.83%)
Some other race	13 (2.17%)
American Indian or Alaskan Native	4 (0.67%)
Prefer not to say	8 (1.33%)

3.2 Meme stimuli and rating tasks

Meme stimuli were collected from social media and contained at least one person and a text caption of 50 words or fewer. Each meme depicted potentially sensitive information about strangers, such as clear photos of their faces, without indication of consent for broader distribution. To ensure a diverse representation of social media content, additional parameters were applied during meme selection. Memes included people of varying ages, genders, and racial and ethnic backgrounds. Furthermore, each meme contained private information ranging in type of IDP-relevant content categories, for instance, medical details, drug usage, personal information (e.g., driver's license), sexual history, and other personal information of the meme target. IDP-relevant categories were developed by analyzing a large photo dataset using a thematic approach [7]. Memes did not include explicitly sexist, racist, or bigoted themes, as radicalized content is not the study focus. While such content is present in real-world social media, our goal was to isolate the effects of peer comments on meme sharing behavior, free from the potential confounding influence of offensive themes. The final selection comprised 13 categories, each represented by five to six memes, for a total of 67 memes; these categories were developed through qualitative thematic analysis of real-world memes [8] and subsequently validated quantitatively [7].

We divided 67 memes into two sets, to evaluate the impact of IDP-relevant peer comment on sharing decisions. 13 memes from various IDP-relevant categories were selected for the '*commented meme-rating task*' to assess the *direct* effects of peer comments. In this case, the memes appeared one-by-one in random order in conjunction with IDP-relevant peer comment that discouraged, encouraged, or offered mixed feedback for meme sharing² (See Figure 5). The remaining 54 memes were presented in the '*meme-rating task*' to assess *carry-over* effects after viewing the peer comment. In this case, only the photo-based memes appeared (no comments)

²Due to oversight, the control group did not rate the 13 memes shown to the experimental group. However, the diverse stimuli and randomized presentation provided a solid foundation for analyzing the effectiveness of peer comment influence in the experimental group. Significant differences between encouraging, discouraging, and mixed conditions highlight varying advocacy levels for peer comments.

to examine whether there was lasting influence from seeing IDP-relevant comment from peer. In both the *commented meme-rating task* and the *meme-rating task*—designed to test *direct* and *carry-over* effects of peer comments—participants rated their likelihood of sharing memes on social media. Responses were recorded on a 5-point scale ranging from 1 (*extremely unlikely*) to 5 (*extremely likely*), with higher average ratings indicating a greater willingness to share potentially privacy-violating content.

3.3 Peer comment selection

In order to balance experimental control with ecological validity, we not only used real-world social media memes, we also tested the peer influence using real social media peer comments. We utilized Google's image search feature [34] to extract comments originally posted on the 13 memes used in the *commented meme-rating task* from various social networks. Before extracting these comments, we applied specific eligibility criteria, such that comments must be: (1) specific to the memes and not generic, (2) not consist solely of emojis or contain links to external websites, (3) under 50 words in length, and (4) reflect an *encouraging* or *discouraging* stance toward the meme. We set the 50-word limit to balance participant engagement and attention [41, 48, 73]. We included comment word count as a covariate in the model to account for its potential influence on meme sharing.

In cases where not enough eligible comments were available, we created fake comments while ensuring they met the specified eligibility criteria. Original comments had an average word count of 20.90, while fake comments averaged 14.38 words. Comments were categorized into two sets based on their tone. *Encouraging comments* promoted the sharing of the given meme, whereas *discouraging comments* expressed disagreement with sharing the meme. For example, in a meme featuring a child's photo, an encouraging comment might be, "Aww! Your daughter looks so adorable," while a discouraging comment could be, "Why do you need to share your daughter's photo on the internet? There are a lot of creeps out there." Mean word counts were $M = 17.29$ ($SD = 8.25$) for encouraging comments and $M = 20.79$ ($SD = 9.28$) for discouraging comments.

We developed these rating tasks using Microsoft Paint, combining memes and comments to simulate a relatively realistic social media experience. To minimize distractions, usernames and profile pictures associated with comments were blurred using Facepixmap [29], allowing participants to focus on the content of the memes and comments. Participants were randomly assigned to one of four between-subject conditions. Those in the experimental groups ($n = 450$) completed both the *commented meme-rating task* and the subsequent *meme-rating task*. Participants in the control group ($n = 150$) completed only the *meme-rating task* and were not shown any comments.

3.4 Procedure

Following informed consent, followed by an eligibility questionnaire and a content warning noting that some memes could include derogatory, sexualized, or stereotypical content. As shown in Figure 1, participants then completed a social media usage questionnaire that asked about their audience preferences and the ownership of shared photos. Participants next completed the *commented*

meme-rating task, viewing memes paired with user comments and rating each meme after reading both. Those in the experimental group proceeded to the *commented meme-rating task*, followed by a *meme-rating task* in which memes were shown without comments. These tasks were analyzed separately: the *commented meme-rating task* measured the immediate impact of peer comments, while the *meme-rating task* assessed carry-over effects once comments were removed.

Afterward, participants completed a meme sharing questionnaire about their sharing practices and beliefs, including reasons for sharing, engagement behaviors (e.g., re-sharing, rating, commenting), and estimates of the proportion of memes they believed violated the depicted individuals' privacy (0–100% in 5 categories). They also rated the extent to which memes should be considered the property of the individuals portrayed (1 = *Strongly disagree*, 5 = *Strongly agree*). Participants then completed a privacy violation questionnaire with three yes/no items: whether they had requested a photo removal, been asked to remove a photo or tag, or had their comments deleted or reported. Next, they provided an open-ended response describing their primary concerns about social media use. Finally, they answered demographic questions about race and gender identity (self-identified as man, woman, or non-binary). The full questionnaire is provided in Appendix A. On average, the study took 25 minutes to complete.

3.5 Meme valence ratings

We conducted a separate study to collect meme valence ratings, involving an independent group of participants who assessed the perceived valence (i.e., how positively or negatively the meme target was portrayed) of the 67 memes described in Section 3.2. Participants were recruited via Amazon's Mechanical Turk platform. To qualify, participants had to reside in the United States, be fluent in English, aged 18 to 60, and regular social media users with an active social media account (i.e., individuals who visited at least once per week). A total of 104 participants were presented with each meme in a random sequence and asked to rate the valence of each meme based on the following criterion: "To what extent does this post portray the person in the photo positively or negatively?". Participants' responses were recorded on a 5-point scale (1 = *very negative*, 5 = *very positive*), with higher scores reflecting more positive and lower scores more negative portrayals of meme targets. Valence ratings were collected independently from the main study, and the ratings obtained were not shared with participants in the current study.

4 Results

4.1 Descriptive statistics

This section summarizes participants' social media use and photo-sharing behaviors. Sharing practices varied across participants: 57.17% typically shared or re-shared photos with friends, 6.83% shared publicly, 30% shared both publicly and with friends, and 6% reported never sharing photos. With respect to photo ownership, 39.67% of participants reported re-sharing others' photos, 36.33% shared their own photos, 17.83% engaged in both behaviors, and 6% did neither. Beyond these general practices, Table 3 summarizes participants' meme-sharing behaviors, including sharing

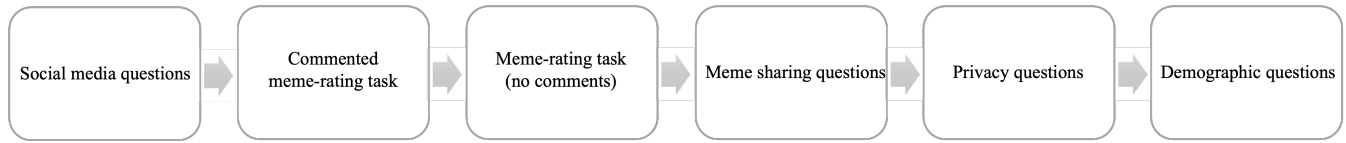


Figure 1: Workflow of the survey

motivations, engagement with liked memes, and perceptions of privacy-violating content. Table 4 reports participants' experiences with privacy violations on social media, such as photo removal requests and comment deletion or reporting.

Building on prior work linking gender to IDP-relevant sharing frequency [7], we next examined whether sharing likelihood differed by gender across the *commented meme-rating task* and the *meme-rating task*. For each task, we calculated average sharing likelihood and conducted one-way ANOVAs to assess gender differences. In the *commented meme-rating task*, sharing likelihood varied significantly by gender ($F(2, 447) = 5.17, p < 0.01$), with men ($M = 1.91, SD = 0.73$) reporting higher sharing likelihood than women ($M = 1.70, SD = 0.60$). A similar pattern emerged in the *meme-rating task*, where men ($M = 1.95, SD = 0.83$) again reported greater sharing likelihood than women ($M = 1.74, SD = 0.67$), $F(2, 597) = 5.77, p < 0.01$. No significant differences were observed for non-binary participants ($p > 0.05$). Given these consistent differences across tasks, gender was included as a covariate in subsequent analyses.

4.2 Model specification

Our analyses used meme sharing likelihood ratings (1 = *extremely unlikely*, 5 = *extremely likely*) as the outcome. The primary predictor was *peer influence type* (between-subjects: encouraging, discouraging, mixed, control). We analyzed two predefined scenarios based on task structure: the *commented meme-rating task* (direct effects while comments were visible; experimental conditions only) and the *meme-rating task* (carry-over effects after comments were removed; experimental and control participants). Where indicated, we included two moderators: *meme valence* (1 = *very negative*, 5 = *very positive*) and *past online misconduct* (participant-level binary indicator: whether one's comments had ever been deleted/reported). Because each participant provided multiple meme ratings, observations were not independent. We therefore used linear mixed-effects models with random intercepts for participants to account for within-participant correlation and stable individual differences in baseline sharing propensity, while estimating population-level effects of peer influence and other predictors. Gender was included as a covariate in all models due to baseline differences in sharing likelihood as explained in 4.1. Comment word count was included only in models for the *commented meme-rating task*, where comment length varies and can affect engagement as explained in 3.3, and was omitted in the *meme-rating task* (no comments), where it was not applicable. Across models, fixed effects were introduced incrementally to examine (1) the main effects of peer influence (M1-M2), (2) moderation by meme valence (M3-M4), and (3) moderation by past online misconduct (M5-M6). Full model specifications are reported in Appendix A.

4.3 Peer comments and the persistence of their influence on sharing decisions

To address RQ1 and RQ2, we tested how peer comments influenced meme sharing and whether these effects persisted over time. For RQ1, we hypothesized that encouraging comments would increase sharing, while discouraging comments would reduce it (H1). We ran a linear mixed-effects regression with peer influence type (encouraging, discouraging, mixed) as the predictor and meme sharing likelihood during the *commented meme-rating task* as the outcome. We accounted for random variability across participants and included participants' gender and comment word count as covariates to control for their influence on sharing likelihood. Results showed a significant main effect of peer influence ($p < 0.05$; Table 5 (M1), Appendix A). Specifically, discouraging comments reduced sharing relative to encouraging comments ($\beta = -0.139, p < 0.05$), whereas mixed comments increased sharing relative to encouraging comments ($\beta = 0.246, p < 0.001$) and to discouraging comments ($\beta = 0.385, p < 0.001$). Gender was a significant predictor: men shared more than women ($\beta = 0.167, p < 0.01$), while non-binary participants did not differ significantly from others ($p > 0.05$). Longer comments were also associated with reduced sharing likelihood ($\beta = -0.212, p < 0.001$).

For RQ2 (H2: persistence of peer comment influence), we examined whether the effects of peer comments carried over into subsequent sharing decisions by estimating a second linear mixed-effects model (M2). In this model, peer influence type was included as the predictor and meme sharing likelihood in the *meme-rating task* served as the outcome. Because no peer comments were provided in this follow-up task, only participants' gender was included as a covariate to account for its influence on sharing behavior. Results indicated no significant effects of peer influence on later sharing ($p > 0.05$; Table 5 (M2)), although men again shared more than women ($\beta = 0.170, p = 0.001$). The lack of significant peer comment effects suggests that initial impacts on meme sharing attenuated over time, providing no support for H2. We revisit this hypothesis in Sections 4.4 and 4.5 for a more nuanced analysis.

4.4 Photo target portrayal impacts peer comment influence

RQ3 examined whether the portrayal of meme targets shaped the effectiveness of peer comment influence. To test this, we ran a linear mixed-effects model (M3) that predicted sharing likelihood in the *commented meme-rating task*. We included peer influence type and meme valence as predictors, modeled participant as a random effect, and controlled for gender and comment word count as covariates.

Results showed a significant main effect of peer comment influence ($p < 0.05$; Table 6 (M3)). Specifically, mixed comments increased sharing relative to both encouraging comments ($\beta = 0.155$,

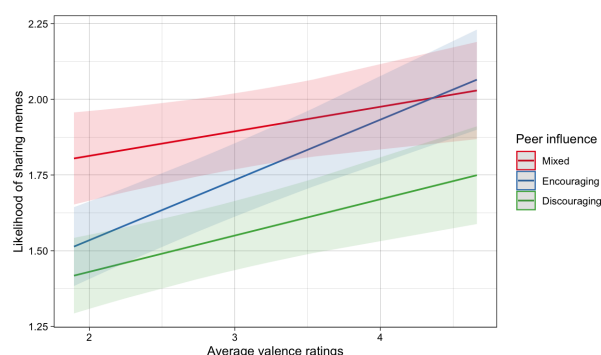


Figure 2: Interaction effect of average meme valence on commented meme-sharing task

$p = 0.001$), and discouraging comments ($\beta = 0.304$, $p < 0.01$). In contrast, discouraging comments did not differ significantly compared to encouraging comments ($p > 0.05$). There was also a significant main effect of meme valence ($\beta = 0.129$, $p < 0.001$), indicating that memes with more positive portrayals were more likely to be shared. Furthermore, men reported a greater likelihood of sharing than women ($\beta = 0.167$, $p = 0.003$), and longer comments were associated with lower sharing likelihood ($\beta = -0.151$, $p < 0.001$). We also found a significant interaction between peer influence type and meme valence ($p < 0.05$). In Figure 2, the x-axis represents meme valence on a scale from 1 to 5, where higher values indicate more positive portrayals of the meme target and lower values indicate more negative portrayals, and the y-axis represents sharing likelihood on a scale from 1 to 5. As shown in Figure 2, the likelihood of sharing increased as meme portrayals became more positive. We observed a significant positive interaction between encouraging comments and meme valence relative to mixed comments ($\beta = 0.077$, $p = 0.005$), indicating that as portrayals of the meme target became more positive, participants receiving encouraging comments were significantly more likely to share the meme. This association was significantly attenuated under mixed comments compared to encouraging comments ($\beta = -0.077$, $p = 0.005$). The interaction between discouraging comments and meme valence was also attenuated relative to encouraging comments, although this effect was only marginally significant ($\beta = -0.051$, $p = 0.054$). These results suggest that encouraging comments amplified the impact of positive portrayals on sharing more than mixed or discouraging comments. Conversely, participants exposed to mixed comments were relatively more likely to share memes with negative portrayals than those exposed to encouraging comments.

We also examined whether peer comment effects carried over to the subsequent *meme-rating task* (no comments) using a second model (M4) with the same predictors as M2. Peer influence type remained significant ($p < 0.05$; Table 6 (M4)). Follow-up comparisons indicated that discouraging comments reduced sharing relative to the control group ($\beta = -0.006$, $p < 0.05$). In contrast, neither the encouraging nor the mixed condition differed significantly from the control group ($p > 0.05$). Meme valence continued to positively

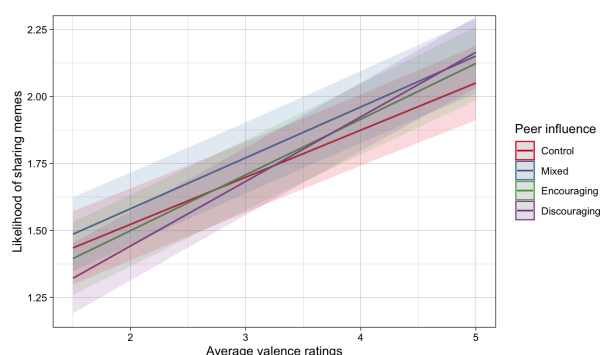


Figure 3: Interaction effect of average meme valence on meme-sharing task (no comments)

predict sharing ($\beta = 0.126$, $p < 0.001$), and gender remained significant with men reporting higher sharing likelihood than women ($\beta = 0.170$, $p = 0.001$). The interaction between peer influence type and meme valence was significant ($p < 0.05$). Figure 3 illustrates this interaction, with meme valence on the x-axis (1 = *negative portrayal*, 5 = *positive portrayal*) and meme sharing likelihood on the y-axis (1-5). The control group serves as the reference level for all comparisons. As shown in Figure 3, discouraging comments most strongly reduced the sharing of negatively portrayed memes ($\beta = 0.047$, $p < 0.001$). Encouraging comments had a marginal positive effect for positively portrayed memes ($\beta = 0.023$, $p = 0.059$), while mixed comments showed no significant interaction ($p > 0.05$). Overall, meme valence consistently shaped sharing decisions, with discouraging comments most effective in reducing the spread of negatively portrayed memes even after peer comments were no longer visible.

4.5 Impact of individual differences

To answer RQ4, we asked participants whether their comments on other people's posts had ever been deleted or reported by someone on social media. We hypothesized that individuals who self-report past allegations of online misconduct are less responsive to the peer comment influence (H4). In our mixed-effects linear regression model (M5), we analyzed the likelihood of meme sharing during the *commented meme-rating task* as the outcome variable, using peer influence type and past online misconduct (Y/N) as predictors³. Second, new effects emerged, including the positive main effect of past online misconduct on meme sharing likelihood ($\beta = 0.337$, $p = 0.003$); See Table 7 (M5). We also found a marginally significant interaction between mixed comments and past online misconduct on meme sharing likelihood, relative to encouraging comments ($\beta = -0.307$, $p = 0.054$). This effect indicates that people who had

³We reran M5 by adding meme valence as a predictor along with peer influence type and past online misconduct (Y/N), with encouraging comments used as the reference level. Participant's gender and word count of the comment were covariates. First, a number of effects were similar to M5: Valence was a strong main predictor ($\beta = 0.144$, $p < 0.001$), the interaction between mixed comments and valence ($\beta = -0.082$, $p = 0.008$) and the interaction between discouraging comments and valence ($\beta = -0.063$, $p = 0.032$) remained significant, while none of the higher-order interactions with past online misconduct were significant ($p > 0.05$). The key moderation effect of past online misconduct was not statistically significant ($p > 0.05$).

their comments deleted or reported were less likely to share memes under the mixed peer comment condition, compared to people not reporting past online misconduct.

Next, we tested whether past experiences of online misconduct moderated the carry-over effects of peer influence on subsequent meme sharing. We estimated a mixed-effects linear model (M6) for the *meme-rating task* (no comments), using the same predictors and covariates as in the previous models (M2, M4). The main effect of past online misconduct was not significant ($p > 0.05$; Table 7, M6), but past online misconduct significantly moderated the effects of peer comments on meme-sharing likelihood ($p < 0.05$). Specifically, among participants with past online misconduct (i.e., whose comments had previously been deleted or reported), exposure to encouraging ($\beta = 0.437$, $p = 0.013$) or discouraging comments ($\beta = 0.387$, $p = 0.036$) was associated with higher meme-sharing likelihood relative to the control condition. Figure 4 illustrates the interaction between past online misconduct and meme sharing likelihood. The x-axis represents participants' binary responses to whether their comments on others' posts had ever been deleted or reported (yes vs. no) within each experimental condition, where blue indicates "yes" and red indicates "no." The y-axis represents meme sharing likelihood on social media (1-5). Individuals with past online misconduct were more likely to share memes when exposed to encouraging or discouraging comments compared to those without such experiences. All other contrasts were non-significant ($p > 0.05$). Table 2 summarizes the hypothesis testing results across all models.

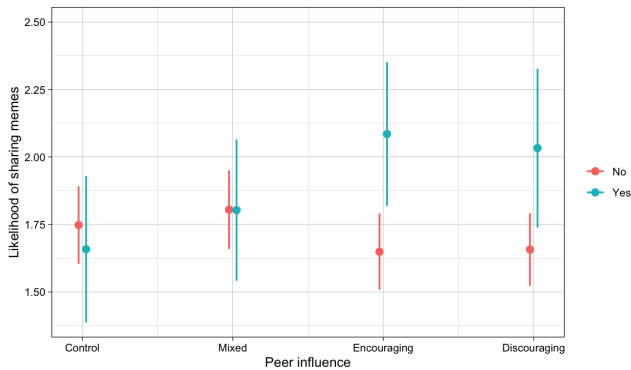


Figure 4: Interaction effect of past online misconduct on meme sharing, with blue and red lines representing participants who answered “Yes” or “No” to having their comments deleted or reported on social media.

5 Discussion

This study examined how peer comments shape privacy-relevant photo sharing. Our results suggest that peer comments do more than momentarily influence decisions: they activate broader social norms around privacy and care. Discouraging comments, in particular, had stronger and more durable effects than encouraging, and mixed feedback. This aligns with social normative influence research, which shows that people adjust behavior in response to

clear signals about group norms [50]. Unlike encouraging or mixed responses, discouraging comments highlight ethical or normative violations, making them especially salient as negative reinforcement [10]. Findings show the ways in which users can support an ethics of care in social media, where an ethics of care framework emphasizes relational responsibility and concern for others' well-being as guides to ethical decision-making [6, 30]. Comments that discourage sharing of potentially private information are effective in supporting a network of care. Not only is such feedback effective in the short-term, it persists even after comments are no longer visible because users internalize the embedded relational and normative expectations. Yet this dynamic does not extend to all forms of feedback: mixed comments, even when well-intentioned, had the opposite effect by amplifying sharing. While this pattern is consistent with a conflict between competing social norms, it may also reflect other cognitive processes, such as attentional imbalance, familiarity with encouragement norms, or differential weighting of positive versus negative cues. When encouragement and discouragement appeared together, the encouragement tended to outweigh the discouragement, consistent with research on positivity bias [59]. One reason may be that social media users are habituated to seeing encouragement for sharing humorous or meme-like content, which is a normative practice, whereas discouraging the sharing of content without consent is less common. In this context, signals of encouragement may have felt more familiar and persuasive, making efforts to discourage sharing not simply neutralized but producing a net-negative effect on privacy.

A small number of brief comments produced lasting (albeit short-term) carry-over effects on privacy-relevant sharing. Our findings suggest that users recalibrate privacy standards quickly in response to micro-interactions, highlighting the value of fostering a culture of responsible sharing that acknowledges broader consequences—such as misinformation, cyberbullying, and privacy violations—without discouraging creativity or engagement. At the same time, peer comments are not a ‘one-size-fits-all’ solution. Our study shows that past online misconduct shapes responses to peer feedback: for individuals with such experiences, both encouraging and discouraging comments unexpectedly increased sharing. These users may be desensitized to peer influence or hold conflicting privacy attitudes [20, 66]. A history of misconduct may signal ‘hardened’ privacy attitudes that are less responsive to community norms [17, 39].

Together, these findings have implications for understanding social influence and the design of feedback mechanisms on the Web. Socio-technical systems such as social media amplify user interactions through personalization and curation, which can reinforce filter bubbles and distort collective norms [18, 60, 67]. Our results show that even lightweight peer interactions can recalibrate behavior and potentially counteract these distortions. Rather than relying on top-down enforcement by platforms or institutions, our findings point to the value of bottom-up approaches that empower users to engage in collective privacy management. Peer-driven feedback offer a scalable, community-based mechanism for strengthening collective norms: by surfacing feedback that emphasizes ethical responsibility, Web users themselves can regulate sharing practices, prioritize the protection of others' privacy, and reinforce collective accountability in online communities.

Table 2: Hypotheses testing results

Hypotheses	Supported
H1: Encouraging comments will increase meme sharing likelihood, while discouraging comments will reduce it	Partially
H2: Peer comments will exhibit carry-over effects, influencing subsequent meme sharing decisions	Partially
H3: Peer comment influence will be more effective at reducing meme sharing with negatively valenced content	Yes
H4: Peer comment influence will be less effective for individuals with past online misconduct	Yes

6 Design implications

Our findings highlight the potential of peer comment interactions as lightweight, bottom-up mechanisms for promoting IDP on the Web. Rather than depending solely on platform-driven moderation or regulatory enforcement, our results show that peer feedback—especially discouraging comments—can significantly reduce the sharing of privacy-sensitive content. This points to the value of designing systems that foreground peer advocacy and collective accountability, aligning with Web research on socio-technical systems, online self-governance, and community-based norm enforcement. Yet, our results also show that mixed comments increased sharing, illustrating how conflicting privacy standards can embolden users to disclose more rather than less. This finding underscores that comments and nudges cannot be assumed to have uniformly positive or neutral effects; in some contexts, they may encourage the very behaviors they are intended to counteract, highlighting the importance of careful design and contextual sensitivity.

One promising direction is to make peer influence more visible and intentional. Current mechanisms such as reporting often remove content without communicating shared norms to the broader community. By contrast, design features could highlight when peers express concerns about privacy risks, allowing such comments to function as social signals at the moment of sharing. For example, when a user drafts a post that includes sensitive personal details, the interface could surface anonymized snippets of prior peer feedback (e.g., “users in your network often discourage sharing when private information is visible”). This approach emphasizes normative guidance from peers rather than automated censorship, strengthening collective privacy management.

Our results also suggest the need for tools that facilitate respectful communication of privacy preferences. We observed that even casual expressions of discouragement shifted sharing behavior, pointing to the role of *privacy gestures* in online interactions. To build on this, social media platforms could offer *Customizable Privacy Templates* that allow users to embed privacy preferences into their shared content. For example, a ‘Family Photos’ template might blur faces and restrict visibility to close contacts, while an ‘Event Photos’ template could allow broader sharing but remove location data. These templates would not only protect sensitive information but also signal collective expectations around appropriate sharing, making unauthorized disclosures less socially acceptable.

7 Limitations and future work

One limitation of our study is its reliance on self-report data, where responses may not accurately reflect real-world behaviors. For example, participants could have adjusted their responses to align with perceived social norms during the rating tasks. Moreover, our

focus on photo-based meme sharing behavior may not capture the full complexity of other types of content shared on social media. Although we worked to standardize memes and peer comments used in the study, they cannot represent the full spectrum of information encountered online. Nonetheless, this research marks an initial step toward understanding how user feedback can promote responsible sharing practices on social media. Lastly, while our study captures immediate responses, these may not reflect long-term changes. Future work can explore more complex peer interactions, for instance, using real-life stories as peer feedback (e.g., to examine the effect of emotions on sharing decisions).

8 Conclusion

Our study demonstrates that peer comments can shape IDP-relevant sharing on the Web, revealing how privacy norms are formed, reinforced, and contested within socio-technical systems. Rather than treating privacy as only an individual choice or platform responsibility, our findings foreground collective privacy management: the ways in which users negotiate responsibility for one another’s well-being. This perspective challenges approaches that rely primarily on automated moderation, and instead points to the power of everyday interactions such as comments, gestures, and feedback as sites where ethics of care are enacted. For Web research, this shift raises new questions: How can systems amplify constructive peer influence without spreading bias or exclusion? What does online self-governance look like when collective responsibility is distributed across diverse communities? And how might peer-driven mechanisms complement, rather than compete with, formal privacy education and governance? We invite researchers and designers to view IDP not as an individual burden but as a shared social practice, and to imagine design strategies that foster more accountable, empathetic, and resilient online communities. At the same time, our findings point to the limits of peer comments and nudges: while they are potentially powerful tools at scale, they can also backfire among certain user subgroups (e.g., those with past online misconduct) or in contexts with conflicting social norms. In such cases, educational approaches that explicitly surface and reconcile normative tensions may be more beneficial than nudges [7, 70].

Acknowledgments

This material is based upon work supported in part by the National Science Foundation under grant CNS-2053152 and by the University of Central Florida (UCF). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the sponsors.

References

- [1] Alessandro Acquisti, Idris Adjerid, Rebecca Balebako, Laura Brandimarte, Lorie Faith Cranor, Saranga Komanduri, Pedro Giovanni Leon, Norman Sadeh, Florian Schaub, Manya Sleeper, et al. 2017. Nudges for privacy and security: Understanding and assisting users' choices online. *ACM Computing Surveys (CSUR)* 50, 3 (2017), 1–41.
- [2] Alessandro Acquisti and Ralph Gross. 2006. Imagined communities: Awareness, information sharing, and privacy on the Facebook. In *International workshop on privacy enhancing technologies*. Springer, 36–58.
- [3] A Acquisti and R Gross. 2018. Awareness, information sharing, and privacy on the Facebook. In *Proceedings of 6th Workshop on Privacy Enhancing Technologies*. 36–58.
- [4] Tala Almaghrabi. 2023. *Memes in Marketing: Exploring how internet memes can be integrated into social media marketing activities in a manner that is accepted by online communities*. Ph.D. Dissertation. Brunel University London.
- [5] Heba Aly, Yizhou Liu, Reza Ghaiumy Anaraki, Sushmita Khan, Moses Namara, Kaileigh Angela Byrne, and Bart Knijnenburg. 2024. Tailoring Digital Privacy Education Interventions for Older Adults: A Comparative Study on Modality Preferences and Effectiveness. *Proceedings on Privacy Enhancing Technologies* (2024).
- [6] Mary Jean Amon. 2022. Promoting interdependent privacy preservation in social media via an ethics of care. In *Privacy Enhancing Technologies Symposium: Workshop on Interdependent and Multi-party Privacy*.
- [7] Mary Jean Amon, Rakibul Hasan, Kurt Hugenberg, Bennett I Bertenthal, and Apu Kapadia. 2020. Influencing photo sharing decisions on social media: A case of paradoxical findings. In *2020 IEEE Symposium on Security and Privacy (SP)*. IEEE, 1350–1366.
- [8] Mary Jean Amon, Aaron Necaise, Nika Kartvelishvili, Aneka Williams, Yan Solihin, and Apu Kapadia. 2023. Modeling user characteristics associated with interdependent privacy perceptions on social media. *ACM Transactions on Computer-Human Interaction* 30, 3 (2023), 1–32.
- [9] Sinan Aral and Dylan Walker. 2012. Identifying influential and susceptible members of social networks. *Science* 337, 6092 (2012), 337–341.
- [10] Roy F Baumeister and Mark R Leary. 2017. The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Interpersonal development* (2017), 57–89.
- [11] Beth T Bell, Jennifer A Cassarly, and Lucy Dunbar. 2018. Selfie-objectification: Self-objectification and positive feedback (“likes”) are associated with frequency of posting sexually objectifying self-images on social media. *Body image* 26 (2018), 83–89.
- [12] Alexis Benveniste. 2022. The Meaning and History of Memes. <https://www.nytimes.com/2022/01/26/crosswords/what-is-a-meme.html>. The New York Times.
- [13] Gergely Biczók and Pern Hui Chia. 2013. Interdependent privacy: Let me share your data. In *Financial Cryptography and Data Security: 17th International Conference, FC 2013, Okinawa, Japan, April 1–5, 2013, Revised Selected Papers* 17. Springer, 338–353.
- [14] Robert M Bond, Christopher J Fariss, Jason J Jones, Adam DI Kramer, Cameron Marlow, Jaime E Settle, and James H Fowler. 2012. A 61-million-person experiment in social influence and political mobilization. *Nature* 489, 7415 (2012), 295–298.
- [15] Danah Michele Boyd. 2008. *Taken out of context: American teen sociality in networked publics*. University of California, Berkeley.
- [16] Helen Brown. 2022. The Surprising Power of Internet Memes. In *Online Resource*. Read article on BBC Future, accessed on 2025-02-03.
- [17] Ricardo Buettner. 2015. Analyzing the problem of employee internal social network site avoidance: Are users resistant due to their privacy concerns?. In *2015 48th Hawaii International Conference on System Sciences*. IEEE, 1819–1828.
- [18] Kelly Carskadon. n.d.. Trying to Find Your Facebook Comments? What to Know About Comment Ranking. In *Online Resource*. Read online, accessed on 2025-02-03.
- [19] Luiz Henrique CB Cavalcanti, Alita Pinto, Jed R Brubaker, and Lynn S Domrowski. 2017. Media, meaning, and context loss in ephemeral communication platforms: A qualitative investigation on Snapchat. In *Proceedings of the 2017 ACM conference on computer supported cooperative work and social computing*. 1934–1945.
- [20] Eshwar Chandrasekharan, Shagun Jhaver, Amy Bruckman, and Eric Gilbert. 2022. Quarantined! Examining the effects of a community-wide moderation intervention on Reddit. *ACM Transactions on Computer-Human Interaction (TOCHI)* 29, 4 (2022), 1–26.
- [21] Jin Chen, Jerry Wenjie Ping, Yunjie Xu, and Bernard CY Tan. 2015. Information privacy concern about peer disclosure in online social networks. *IEEE Transactions on Engineering Management* 62, 3 (2015), 311–324.
- [22] Shu-Chuan Chu and Sejung Marina Choi. 2011. Electronic word-of-mouth in social networking sites: A cross-cultural study of the United States and China. *Journal of Global Marketing* 24, 3 (2011), 263–281.
- [23] Tharntip Chutikulrunsee and Oliver Burmeister. 2017. Interdependent privacy. In *CEPE/ETHICOMP Conference 2017*. Elsevier, 1–14.
- [24] Danielle Keats Citron and Daniel J Solove. 2022. Privacy harms. *BUL Rev* 102 (2022), 793.
- [25] Abhishek Das, Japsimar Singh Wahi, and Siyao Li. 2020. Detecting hate speech in multi-modal memes. *arXiv preprint arXiv:2012.14891* (2020).
- [26] Richard Dawkins. 2016. *The selfish gene*. Oxford university press.
- [27] Julia R DeCook. 2018. Memes and symbolic violence: #proudboys and the use of memes for propaganda and the construction of collective identity. *Learning, Media and Technology* 43, 4 (2018), 485–504.
- [28] Dr. Paula Dootson. 2020. 3.2 Billion Images and 720,000 Hours of Video Are Shared Online Daily. Can You Sort Real from Fake?. In *Online Resource*. Read online, accessed on 2025-02-03.
- [29] Facepixelizer. n.d.. Facepixelizer. In *Online Resource*. <https://www.facepixelizer.com/>, accessed on 2025-02-03.
- [30] Marilyn Friedman. 2015. Privacy, surveillance, and care ethics. *Care ethics and political theory* (2015), 108–126.
- [31] Siyao Fu, Haibo He, and Zeng-Guang Hou. 2014. Learning race from face: A survey. *IEEE transactions on pattern analysis and machine intelligence* 36, 12 (2014), 2483–2509.
- [32] Sherice Gearhart and Weiwu Zhang. 2015. “Was it something I said?” “No, it was something you posted!” A study of the spiral of silence theory in social media contexts. *Cyberpsychology, Behavior, and Social Networking* 18, 4 (2015), 208–213.
- [33] Kambiz Ghazinoor and John Ponchak. 2017. Hidden privacy risks in sharing pictures on social media. *Procedia computer science* 113 (2017), 267–272.
- [34] Google. n.d.. Google Image Search. In *Online Resource*. <https://images.google.com/>, accessed on 2025-02-03.
- [35] Hana Habib, Neil Shah, and Rajan Vaish. 2019. Impact of contextual factors on snapchat public sharing. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [36] Leslie A Hahner. 2013. The riot kiss: Framing memes as visual argument. *Argumentation and advocacy* 49, 3 (2013), 151–166.
- [37] Mathias Humbert, Benjamin Trubert, and Kévin Huguenin. 2019. A survey on interdependent privacy. *ACM Computing Surveys (CSUR)* 52, 6 (2019), 1–40.
- [38] Jasser Jasser. 2019. Dynamics of misinformation cascades. In *Companion Proceedings of The 2019 World Wide Web Conference*. 33–36.
- [39] Matthew Katsaros, Tom Tyler, Jisu Kim, and Tracey Meares. 2022. Procedural justice and self governance on Twitter: Unpacking the experience of rule breaking on Twitter. *Journal of Online Trust and Safety* 1, 3 (2022).
- [40] Hanna Krasnova and Natasha F Veltri. 2010. Privacy calculus on social networking sites: Explorative evidence from Germany and USA. In *2010 43rd Hawaii international conference on system sciences*. IEEE, 1–10.
- [41] Jon A Krosnick. 1991. Response strategies for coping with the cognitive demands of attitude measures in surveys. *Applied cognitive psychology* 5, 3 (1991), 213–236.
- [42] Anne Leiser. 2022. Psychological perspectives on participatory culture: Core motives for the use of political internet memes. *Journal of Social and Political Psychology* 10, 1 (2022), 236–252.
- [43] Amanda Lenhart, Kristen Purcell, Aaron Smith, and Kathryn Zickuhr. 2010. Social media & mobile internet use among teens and young adults. *Millennials* (2010).
- [44] Jie Li, Yiping Kong, Thomas Röggla, Francesca De Simone, Swamy Ananthanarayan, Huib De Ridder, Abdallah El Ali, and Pablo Cesar. 2019. Measuring and understanding photo sharing experiences in social virtual reality. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [45] Pengxiang Li, Leanne Chang, Trudy Hui Hui Chua, and Renae Sze Ming Loh. 2018. “Likes” as KPI: An examination of teenage girls’ perspective on peer feedback on Instagram and its influence on coping response. *Telematics and Informatics* 35, 7 (2018), 1994–2005.
- [46] Charles F Manski. 1993. Identification of endogenous social effects: The reflection problem. *The review of economic studies* 60, 3 (1993), 531–542.
- [47] Michael Massimi, Khai Truong, David Dearman, and Gillian Hayes. 2009. Understanding recording technologies in everyday life. *IEEE Pervasive Computing* 9, 3 (2009), 64–71.
- [48] Stacey McLachlan and Brayden Cohen. 2021. Experiment: Do Long Captions Get More Engagement on Instagram?. In *Online Resource*. <https://blog.hootsuite.com/long-instagram-captions-experiment/>, accessed on 2025-02-03.
- [49] Sarah McRoberts, Haiwei Ma, Andrew Hall, and Svetlana Yarosh. 2017. Share first, save later: Performance of self through Snapchat stories. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 6902–6911.
- [50] Erin L Mead, Rajiv N Rimal, Roberta Ferrence, and Joanna E Cohen. 2014. Understanding the sources of normative influence on behavior: The example of tobacco. *Social science & medicine* 115 (2014), 139–143.
- [51] Kevin Munger. 2017. Tweetment effects on the tweeted: Experimentally reducing racist harassment. *Political Behavior* 39, 3 (2017), 629–649.
- [52] Marek Muszyński. 2023. Attention checks and how to use them: Review and practical recommendations. (2023).
- [53] Aaron Necaise, Tangila Islam Tanni, Aneka Williams, Yan Solihin, Apu Kapadia, and Mary Jean Amon. 2023. User Preferences for Interdependent Privacy Preservation Strategies in Social Media. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW2 (2023), 1–30.

- [54] BBC News. 2015. 'Can't hide it forever': The model who became a meme. In *Online Resource*. <https://www.bbc.com/news/world-asia-34568674>, accessed on 2025-02-03.
- [55] Helen Nissenbaum. 2009. Privacy in context: Technology, policy, and the integrity of social life. In *Privacy in Context*. Stanford University Press.
- [56] Rebecca Panskus, Tangila Islam Tanni, Alexander Ponticello, Echo Meissner, Yan Solihin, Katharina Krombholz, and Karola Marky. 2024. Let me quickly share it-Time Pressure when Sharing on Social Media. In *Proceedings of the International Conference on Mobile and Ubiquitous Multimedia*. 280–299.
- [57] Eyal Peer, Serge Egelman, Marian Harbach, Nathan Malkin, Arunesh Mathur, and Alisa Frik. 2020. Nudge me right: Personalizing online security nudges to people's decision-making styles. *Computers in Human Behavior* 109 (2020), 106347.
- [58] Richard Rogers and Giulia Giorgi. 2024. What is a meme, technically speaking? *Information, Communication & Society* 27, 1 (2024), 73–91.
- [59] Paul Rozin and Edward B. Royzman. 2001. Negativity bias, negativity dominance, and contagion. *Personality and Social Psychology Review* 5, 4 (2001), 296–320.
- [60] Philip Seargeant and Caroline Tagg. 2019. Social media and the future of open debate: A user-oriented approach to Facebook's filter bubble conundrum. *Discourse, Context & Media* 27 (2019), 41–48.
- [61] Harshit Sharma. 2018. Memes in digital culture and their role in marketing and communication: A study in India. *Interactions: Studies in Communication & Culture* 9, 3 (2018), 303–318.
- [62] Pavica Sheldon and Katherine Bryant. 2016. Instagram: Motives for its use and relationship to narcissism and contextual age. *Computers in human Behavior* 58 (2016), 89–97.
- [63] Garrett Smith, Kirsten Chapman, Zainab Agha, Janet Ruppert, Spring Cullen, Sushmita Khan, Bart Knijnenburg, Jessica Vitak, Priya C Kumar, Pamela J Wisniewski, et al. 2023. Privacy Interventions and Education (PIE): Encouraging Privacy Protective Behavioral Change Online. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–5.
- [64] Kseniya Stsiampkouskaya, Adam Joinson, Lukasz Piwek, and Carl-Philip Ahlbom. 2021. Emotional responses to likes and comments regulate posting frequency and content change behaviour on social media: An experimental study and mediation model. *Computers in Human Behavior* 124 (2021), 106940.
- [65] Jose M Such, Joel Porter, Sören Preibusch, and Adam Joinson. 2017. Photo privacy conflicts in social media: A large-scale empirical study. In *Proceedings of the 2017 CHI conference on human factors in computing systems*. 3821–3832.
- [66] Jie Tang, Umair Akram, and Wenjing Shi. 2021. Why people need privacy? The role of privacy fatigue in app users' intention to disclose privacy: based on personality traits. *Journal of Enterprise Information Management* 34, 4 (2021), 1097–1120.
- [67] Tangila Islam Tanni, Aaron Necaie, Yan Solihin, and Mary Jean Amon. 2024. Empirical Evidence for Reciprocal Radicalization: Conformity of User Comment, Peer Feedback, and Community Emotion in Extreme Misogynistic Social Media Groups. In *International Conference on Human-Computer Interaction*. Springer, 403–416.
- [68] Martijn Ten Bhömer, John Helmes, Kenton O'Hara, and Elise Van Den Hoven. 2010. 4Photos: a collaborative photo sharing experience. In *Proceedings of the 6th Nordic Conference on Human-Computer Interaction: Extending Boundaries*. 52–61.
- [69] Yang Wang, Gregory Norcie, and Lorrie Faith-Cranor. 2017. Who is concerned about what? a study of american, chinese and indian users' privacy concerns on social network sites (short paper).
- [70] Renita Washburn, Tangila Islam Tanni, Yan Solihin, Apu Kapadia, and Mary Jean Amon. 2023. Bottom-up psychosocial interventions for interdependent privacy: Effectiveness based on individual and content differences. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–20.
- [71] Simon Weaver. 2011. Jokes, rhetoric and embodied racism: A rhetorical discourse analysis of the logics of racist jokes on the internet. *Ethnicities* 11, 4 (2011), 413–435.
- [72] Aneka Williams, Grace Fox, Mary Jean Amon, Tangila Islam Tanni, and Yan Solihin. 2025. The GenAI networked privacy problem at work-How privacy knowledge and perceptions predict Generative AI disclosure in professional contexts. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–9.
- [73] Ting Yan and Roger Tourangeau. 2008. Fast times and easy questions: The effects of age, experience and question complexity on web survey response times. *Applied Cognitive Psychology: The Official Journal of the Society for Applied Research in Memory and Cognition* 22, 1 (2008), 51–68.
- [74] Savvas Zannettou, Tristan Caulfield, Jeremy Blackburn, Emiliano De Cristofaro, Michael Sirivianos, Gianluca Stringhini, and Guillermo Suarez-Tangil. 2018. On the origins of memes by means of fringe web communities. In *Proceedings of the internet measurement conference 2018*. 188–202.
- [75] Anne L Zell and Lisa Moeller. 2018. Are you happy for me... on Facebook? The potential importance of "likes" and comments. *Computers in Human Behavior* 78 (2018), 26–33.

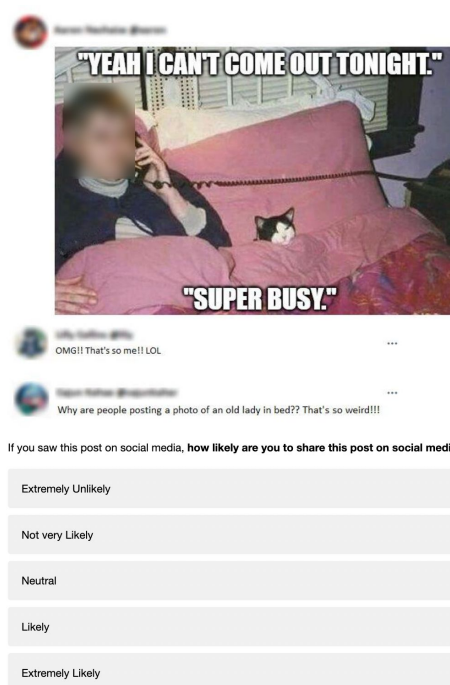
A Supplementary materials

A.1 Social media usage questionnaire

- When you share or re-share photos on social media, who do you typically share them with?
 1. Friends/connections; 2. General viewers/public; 3. Both; 4. NA: I do not share or re-share photos on social media
- Do you share your own photos or re-share other people's photos more often on social media?
 1. I share my own photos more; 2. I re-share other people's photos more; 3. I share and re-share equally; 4. NA: I do not share or re-share photos on social media.

A.2 Social media meme sharing questionnaire

- Please indicate the main reason that you share or re-share memes of other people online.
 1. Humor/entertainment; 2. Personally relate to the content; 3. Connect with others; 4. To fit in/for appearances; 5. Spread awareness; 6. Show support; 7. Self-expression; 8. Store memories/ideas; 9. No reason—just something to do; 10. Other
- How do you respond when you see memes that you like on social media? (Select all that apply.)
 1. Re-share it on my profile; 2. Rate the meme (e.g., up-vote/downvote); 3. Post a comment; 4. Send it to friends/family; 5. None of the above
- In your opinion, what percentage of all the memes presented in this study were potentially violating the privacy of the person shown in the memes?
 1. 0%; 2. 1–25%; 3. 26–50%; 4. 51–75%; 5. 76–100%



If you saw this post on social media, how likely are you to share this post on social media?

Extremely Unlikely

Not very Likely

Neutral

Likely

Extremely Likely

Figure 5: Mixed peer influence rating task

Table 3: Participants' meme sharing behaviors

Measure	Percentage(%)
Motivations for sharing memes	Humor (54.33%); Relatability (21.17%); Self-expression (5.67%); Connection (4.33%); Fitting in (3.33%); Other (11.16%)
Engagement with liked memes	Shared (59.66%); Rated (52.67%); Re-shared (39.00%); Commented (15.00%); None (7.33%)
Perceived proportion of privacy-violating memes	None (3.00%); 1–25% (37.70%); 26–50% (27.00%); 51–75% (21.30%); 76–100% (11.00%)
Memes should be considered the property of depicted individuals	Agree (42.00%); Neutral (31.00%); Disagree (27.00%)

- To what extent do you disagree or agree with the following statement: “Memes should be considered the property of the person(s) portrayed in the meme.”
1. Strongly disagree; 2. Disagree; 3. Neutral; 4. Agree; 5. Strongly agree

A.3 Social media privacy violation questionnaire

- Have you ever asked someone to remove your photo/remove your name from photos that have been tagged to identify you on social media? (Yes/No)

- Has anyone ever asked you to remove their photo/remove their name from photos that have been tagged to identify them on social media? (Yes/No)
- Have your comments on other people's posts been deleted or reported by someone on social media? (Yes/No)
- What is your biggest concern about using social media? Please respond with a minimum of 20 characters. (Open-ended question)

Table 4: Participants' privacy violation experiences

	Encouraging	Discouraging	Mixed	Control
Ever asked someone to remove your photo or untag your name on social media				
Yes	30.67%	38.00%	33.33%	44.00%
No	69.33%	62.00%	66.67%	56.00%
Ever been asked to remove someone else's photo or untag their name on social media				
Yes	16.00%	14.67%	16.00%	19.33%
No	84.00%	85.33%	84.00%	80.67%
Had your comments on others' posts deleted or reported on social media				
Yes	21.33%	16.67%	21.33%	20.00%
No	78.67%	83.33%	78.67%	80.00%

Table 5: Standardized estimates and standard errors for models predicting meme sharing likelihood: commented meme-rating task (M1, direct effects) and meme-rating task (no comments) (M2, carry-over effects)

Predictors	Outcome variable: Sharing likelihood of memes					
	Direct effect on meme sharing (M1)			Carry-over effect on meme sharing (M2)		
	<i>std. β</i>	<i>std. Error</i>	<i>p</i>	<i>std. β</i>	<i>std. Error</i>	<i>p</i>
(Intercept)	-0.100	0.064	<0.001***	-0.088	0.067	<0.001***
Encouraging comments				0.011	0.086	0.883
Discouraging comments	-0.139	0.076	0.036*	-0.006	0.086	0.931
Mixed comments	0.246	0.080	<0.001***	0.062	0.086	0.389
Men	0.167	0.065	0.003**	0.170	0.064	0.001**
Non-binary	0.082	0.152	0.535	0.164	0.152	0.194
Word count of comment	-0.212	0.012	<0.001***			
Observations	5,849			32,400		

Note. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 6: Standardized estimates and standard errors for models predicting meme sharing likelihood based on peer influence type and meme valence: commented meme-rating task (M3) and meme-rating task (M4)

Predictors	Dependent variable: Sharing likelihood of memes					
	Direct effect on meme sharing (M3)			Carry-over effect on meme sharing (M4)		
	<i>std. β</i>	<i>std. Error</i>	<i>p</i>	<i>std. β</i>	<i>std. Error</i>	<i>p</i>
(Intercept)	-0.066	0.115	<0.001***	-0.088	0.077	<0.001***
Encouraging comments				0.011	0.101	0.385
Discouraging comments	-0.150	0.140	0.698	-0.006	0.101	0.038*
Mixed comments	0.155	0.153	0.001**	0.062	0.101	0.765
Meme valence	0.129	0.030	<0.001***	0.126	0.012	<0.001***
Men	0.167	0.065	0.005**	0.170	0.064	0.001**
Non-binary	0.082	0.152	0.535	0.164	0.152	0.194
Word count of comment	-0.151	0.002	<0.001***			
Encouraging comments * Meme valence				0.023	0.017	0.059
Discouraging comments * Meme valence	-0.051	0.041	0.054	0.047	0.017	<0.001***
Mixed comments * Meme valence	-0.077	0.042	0.005**	0.010	0.017	0.409
Observations	5,849			32,400		

Note. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ **Table 7: Standardized estimates and standard errors for models predicting meme sharing likelihood based on peer influence type and past online misconduct: commented meme-rating task (M5) and meme-rating task (M6)**

Predictors	Dependent variable: Sharing likelihood of memes					
	Direct effect on meme sharing (M5)			Carry-over effect on meme sharing (M6)		
	<i>std. β</i>	<i>std. Error</i>	<i>p</i>	<i>std. β</i>	<i>std. Error</i>	<i>p</i>
(Intercept)	-0.164	0.064	<0.001***	-0.065	0.073	<0.001***
Encouraging comments				-0.082	0.096	0.304
Discouraging comments	-0.109	0.083	0.134	-0.076	0.095	0.339
Mixed comments	0.313	0.084	<0.001***	0.047	0.096	0.553
Past online misconduct	0.337	0.129	0.003**	-0.075	0.152	0.554
Men	0.147	0.065	0.009**	0.153	0.063	0.004**
Non-binary	0.055	0.151	0.678	0.150	0.152	0.236
Word count of comment	-0.212	0.001	<0.001***			
Encouraging comments * Past online misconduct				0.437	0.212	0.013*
Discouraging comments * Past online misconduct	-0.099	0.192	0.551	0.387	0.222	0.036*
Mixed comments * Past online misconduct	-0.307	0.183	0.054	0.073	0.211	0.676
Observations	5,849			32,400		

Note. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$